

WEEK-3

3.Extend the cluster to multiple brokers on a single node.

```
cd kafka_2.13-4.3.1/kafka-multi-broker
```

```
java-app/
```

```
└─ pom.xml
```

```
└─ src/main/java/com/example/
```

```
└─ Producer.java
```

```
└─ CreateTopic.java
```

```
└─ Consumer.java
```

docker-compose.yml:-

services:

zookeeper:

image: confluentinc/cp-zookeeper:7.6.1

container_name: zookeeper

ports:

- "2181:2181"

environment:

ZOOKEEPER_CLIENT_PORT: 2181

ZOOKEEPER_TICK_TIME: 2000

networks:

- kafka-net

kafka1:

image: confluentinc/cp-kafka:7.6.1

container_name: kafka1

ports:

- "9092:9092" # inside Docker network
- "29092:29092" # exposed to host

environment:

KAFKA_BROKER_ID: 1

KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181

KAFKA_LISTENER_SECURITY_PROTOCOL_MAP: INSIDE:PLAINTEXT,OUTSIDE:PLAINTEXT

KAFKA_ADVERTISED_LISTENERS: INSIDE://kafka1:9092,OUTSIDE://localhost:29092

KAFKA_LISTENERS: INSIDE://0.0.0.0:9092,OUTSIDE://0.0.0.0:29092

KAFKA_INTER_BROKER_LISTENER_NAME: INSIDE

KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 2

KAFKA_MIN_INSYNC_REPLICAS: 1

depends_on:

- zookeeper

networks:

- kafka-net

kafka2:

image: confluentinc/cp-kafka:7.6.1

container_name: kafka2

ports:

- "9093:9093" # inside Docker network
- "29093:29093" # exposed to host

environment:

KAFKA_BROKER_ID: 2

KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181

KAFKA_LISTENER_SECURITY_PROTOCOL_MAP: INSIDE:PLAINTEXT,OUTSIDE:PLAINTEXT

KAFKA_ADVERTISED_LISTENERS: INSIDE://kafka2:9093,OUTSIDE://localhost:29093

KAFKA_LISTENERS: INSIDE://0.0.0.0:9093,OUTSIDE://0.0.0.0:29093

KAFKA_INTER_BROKER_LISTENER_NAME: INSIDE

KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 2

KAFKA_MIN_INSYNC_REPLICAS: 1

depends_on:

- zookeeper

networks:

- kafka-net

kafka-ui:

image: provectuslabs/kafka-ui:latest

container_name: kafka-ui

ports:

- "8080:8080"

environment:

KAFKA_CLUSTERS_0_NAME: local

KAFKA_CLUSTERS_0_BOOTSTRAPSERVERS: kafka1:9092,kafka2:9093

KAFKA_CLUSTERS_0_ZOOKEEPER: zookeeper:2181

depends_on:

- kafka1
- kafka2

networks:

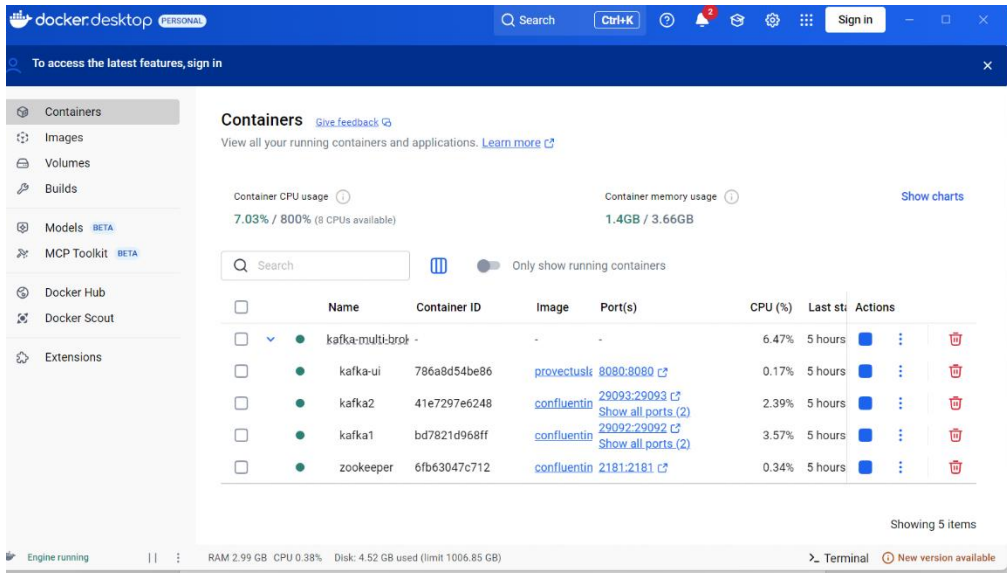
- kafka-net

networks:

kafka-net:

driver: bridge

docker compose up -d



Pom.xml:-

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
```

```
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
```

```
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.example</groupId>
```

```
<artifactId>kafka-java-app</artifactId>
```

```
<version>1.0.0</version>
```

```
<properties>
```

```
    <maven.compiler.source>17</maven.compiler.source>
```

```
    <maven.compiler.target>17</maven.compiler.target>
```

```
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
```

```
</properties>
```

<dependencies>

<dependency>

<groupId>org.apache.kafka</groupId>

<artifactId>kafka-clients</artifactId>

<!-- matches confluentinc/cp-kafka:7.6.1 -> Apache Kafka 3.6.1 -->

<version>3.6.1</version>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.codehaus.mojo</groupId>

<artifactId>exec-maven-plugin</artifactId>

<version>3.1.0</version>

<configuration>

<cleanupDaemonThreads>>false</cleanupDaemonThreads>

</configuration>

</plugin>

</plugins>

</build>

</project>

```
driver: bridge
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker$ ls
docker-compose.yml  java-app
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker$ cd java-app
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$ ls
pom.xml  src  target
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$
```

Producer.java:-

package com.example;

import org.apache.kafka.clients.producer.KafkaProducer;

```
import org.apache.kafka.clients.producer.ProducerRecord;

import org.apache.kafka.clients.producer.RecordMetadata;

import java.util.Properties;

public class Producer {

    public static void main(String[] args) throws Exception {

        Properties props = new Properties();

        props.put(

            "bootstrap.servers",

            "localhost:29092,localhost:29093"

        );

        props.put(

            "key.serializer",

            "org.apache.kafka.common.serialization.StringSerializer"

        );

        props.put(

            "value.serializer",

            "org.apache.kafka.common.serialization.StringSerializer"

        );

        try (KafkaProducer<String, String> producer =

            new KafkaProducer<>(props)) {

            for (int i = 1; i <= 5; i++) {

                String key = "key-" + i;

                String value = "Hello Kafka message " + i;

                ProducerRecord<String, String> record =

                    new ProducerRecord<>(

                        "test-topic",

                        key,
```

```

        value

    );

    RecordMetadata metadata =
        producer.send(record).get();

    System.out.println(
        "Message sent successfully: " +
        "key=" + key +
        ", value=" + value +
        ", partition=" + metadata.partition() +
        ", offset=" + metadata.offset()
    );
}

producer.flush();
} catch (Exception e) {
    System.err.println(
        "Error while producing messages:"
    );
    e.printStackTrace();
}
}
}

```

Consumer.java:-

```

package com.example;

import org.apache.kafka.clients.consumer.ConsumerRecord;

```

```
import org.apache.kafka.clients.consumer.ConsumerRecords;

import org.apache.kafka.clients.consumer.KafkaConsumer;

import java.time.Duration;

import java.util.Collections;

import java.util.Properties;

public class Consumer {

    public static void main(String[] args) {

        Properties props = new Properties();

        props.put("bootstrap.servers", "localhost:29092,localhost:29093");

        props.put("group.id", "test-group");

        props.put(

            "key.deserializer",

            "org.apache.kafka.common.serialization.StringDeserializer"

        );

        props.put(

            "value.deserializer",

            "org.apache.kafka.common.serialization.StringDeserializer"

        );

        props.put("auto.offset.reset", "earliest");

        try (KafkaConsumer<String, String> consumer =

            new KafkaConsumer<>(props)) {

            consumer.subscribe(

                Collections.singletonList("test-topic")

            );

            System.out.println(

                "Listening for messages on test-topic..."

            );

        }

    }

}
```



```

while (true) {

    ConsumerRecords<String, String> records =

        consumer.poll(Duration.ofMillis(1000));

    for (ConsumerRecord<String, String> record : records) {

        System.out.println(

            "Received message: " +

            "key=" + record.key() +

            ", value=" + record.value() +

            ", partition=" + record.partition() +

            ", offset=" + record.offset()

        );

    }

}

} catch (Exception e) {

    System.err.println(

        "Error while consuming messages:"

    );

    e.printStackTrace();

}

}

```

```

nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$ cd src/main/java/com/example
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app/src/main/java/com/example$ ls
Consumer.java  CreateTopic.java  Producer.java
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app/src/main/java/com/example$

```

Createtopic.java:-

```

package com.example;

import org.apache.kafka.clients.admin.AdminClient;

import org.apache.kafka.clients.admin.AdminClientConfig;

import org.apache.kafka.clients.admin.NewTopic;

```

```
import java.util.Collections;

import java.util.Properties;

public class CreateTopic {

    public static void main(String[] args) throws Exception {

        Properties props = new Properties();

        props.put(

            AdminClientConfig.BOOTSTRAP_SERVERS_CONFIG,

            "localhost:29092,localhost:29093"

        );

        try (AdminClient admin = AdminClient.create(props)) {

            NewTopic topic = new NewTopic(

                "test-topic",

                3,

                (short) 2

            );

            admin.createTopics(

                Collections.singleton(topic)

            ).all().get();

            System.out.println(

                "Topic 'test-topic' created successfully."

            );

        } catch (Exception e) {

            System.err.println(

                "Error while creating topic:"

            );

            e.printStackTrace();

        }

    }

}
```

```
}  
  
}
```

```
cd ~/kafka_2.13-4.3.1/kafka-multi-broker/java-app
```

```
mvn clean package
```

```
nagur@LAPTOP-4466380Q:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$ mvn clean package  
[INFO] Scanning for projects...  
[INFO]  
[INFO] -----< com.example:kafka-java-app >-----  
[INFO] Building kafka-java-app 1.0.0  
[INFO] from pom.xml  
[INFO] -----[ jar ]-----  
[INFO]  
[INFO] --- clean:3.2.0:clean (default-clean) @ kafka-java-app ---  
[INFO] Deleting /home/nagur/kafka_2.13-4.3.1/kafka-multi-broker/java-app/target  
[INFO]  
[INFO] --- resources:3.3.1:resources (default-resources) @ kafka-java-app ---  
[INFO] skip non existing resourceDirectory /home/nagur/kafka_2.13-4.3.1/kafka-multi-broker/java-app/src/main/r  
esources  
[INFO]  
[INFO] --- compiler:3.13.0:compile (default-compile) @ kafka-java-app ---  
[INFO] Recompiling the module because of changed source code.  
[INFO] Compiling 3 source files with javac [debug target 17] to target/classes  
[INFO]  
[INFO] --- resources:3.3.1:testResources (default-testResources) @ kafka-java-app ---  
[INFO] skip non existing resourceDirectory /home/nagur/kafka_2.13-4.3.1/kafka-multi-broker/java-app/src/test/r  
esources  
[INFO]  
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ kafka-java-app ---  
[INFO] No sources to compile  
[INFO]  
[INFO] --- surefire:3.2.5:test (default-test) @ kafka-java-app ---  
[INFO]  
[INFO] --- compiler:3.13.0:compile (default-compile) @ kafka-java-app ---  
[INFO] Recompiling the module because of changed source code.  
[INFO] Compiling 3 source files with javac [debug target 17] to target/classes  
[INFO]  
[INFO] --- resources:3.3.1:testResources (default-testResources) @ kafka-java-app ---  
[INFO] skip non existing resourceDirectory /home/nagur/kafka_2.13-4.3.1/kafka-multi-broker/java-app/src/test/r  
esources  
[INFO]  
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ kafka-java-app ---  
[INFO] No sources to compile  
[INFO]  
[INFO] --- surefire:3.2.5:test (default-test) @ kafka-java-app ---  
[INFO] No tests to run.  
[INFO]  
[INFO] --- jar:3.4.1:jar (default-jar) @ kafka-java-app ---  
[INFO] Building jar: /home/nagur/kafka_2.13-4.3.1/kafka-multi-broker/java-app/target/kafka-java-app-1.0.0.jar  
[INFO] -----  
[INFO] BUILD SUCCESS  
[INFO] -----  
[INFO] Total time: 4.255 s  
[INFO] Finished at: 2026-08-11T15:49:39Z  
[INFO] -----
```

mvn exec:java -Dexec.mainClass=com.example.CreateTopic

```
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$ mvn exec:java -Dexec.mainClass=com.example.CreateTopic
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:kafka-java-app >-----
[INFO] Building kafka-java-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ kafka-java-app ---
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Error while creating topic:
java.util.concurrent.ExecutionException: org.apache.kafka.common.errors.TopicExistsException: Topic 'test-topic' already exists.
    at java.base/java.util.concurrent.CompletableFuture.reportGet(CompletableFuture.java:396)
    at java.base/java.util.concurrent.CompletableFuture.get(CompletableFuture.java:2073)
    at org.apache.kafka.common.internals.KafkaFutureImpl.get(KafkaFutureImpl.java:165)
    at com.example.CreateTopic.main(CreateTopic.java:22)
    at org.codehaus.mojo.exec.ExecJavaMojo$1.run(ExecJavaMojo.java:279)
    at java.base/java.lang.Thread.run(Thread.java:840)
Caused by: org.apache.kafka.common.errors.TopicExistsException: Topic 'test-topic' already exists.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 3.139 s
```

mvn exec:java -Dexec.mainClass=com.example.Producer

```
nagur@LAPTOP-446G38UQ: $ cd kafka_2.13-4.3.1/kafka-multi-broker/java-app
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$ mvn exec:java -Dexec.mainClass=com.example.Producer
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:kafka-java-app >-----
[INFO] Building kafka-java-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ kafka-java-app ---
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Message sent successfully: key=key-1, value=Hello Kafka message 1, partition=0, offset=25
Message sent successfully: key=key-2, value=Hello Kafka message 2, partition=0, offset=26
Message sent successfully: key=key-3, value=Hello Kafka message 3, partition=0, offset=27
Message sent successfully: key=key-4, value=Hello Kafka message 4, partition=0, offset=28
Message sent successfully: key=key-5, value=Hello Kafka message 5, partition=0, offset=29
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.453 s
[INFO] Finished at: 2026-08-11T15:51:25Z
[INFO]
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$
```

mvn exec:java -Dexec.mainClass=com.example.Consumer

```
nagur@LAPTOP-446G38UQ: $ cd kafka_2.13-4.3.1/kafka-multi-broker/java-app
nagur@LAPTOP-446G38UQ:~/kafka_2.13-4.3.1/kafka-multi-broker/java-app$ mvn exec:java -Dexec.mainClass=com.example.Consumer
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:kafka-java-app >-----
[INFO] Building kafka-java-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ kafka-java-app ---
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Listening for messages on test-topic...
Received message: key=key-1, value=Hello Kafka message 1, partition=0, offset=25
Received message: key=key-2, value=Hello Kafka message 2, partition=0, offset=26
Received message: key=key-3, value=Hello Kafka message 3, partition=0, offset=27
Received message: key=key-4, value=Hello Kafka message 4, partition=0, offset=28
Received message: key=key-5, value=Hello Kafka message 5, partition=0, offset=29
```